Constructions — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: The two tools*

Key Idea

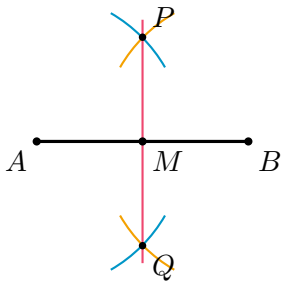
Compass: draws **circles and arcs**, and copies a **length**.

Straightedge: draws straight lines — **no measuring**!

Four basics: copy a segment, copy an angle, bisect a **segment**, bisect an **angle**.

☞ *Video: Bisect a segment (find its midpoint)*

Key Idea



1. From A: open the compass **more than half** of AB. Draw arcs above and below.

2. From B: use the **same** width. Arcs above and below.

3. The arcs cross at P and Q.

4. Line PQ is the **perpendicular bisector**. It meets $\overline{AB}$ at the **midpoint** M.

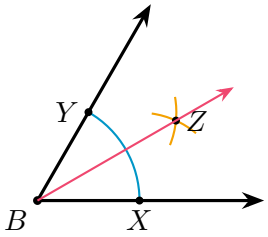
☞ *Video: Watch out*

Common Mistake

Keep the compass width the **same** for both arcs, and open it to **more than half** of AB — otherwise the arcs won’t **cross**.

☞ *Video: Bisect an angle*

Key Idea



1. An arc from the vertex B crosses both sides at X and Y.

2. Same width from X and from Y: the arcs meet at Z.

3. Ray BZ **bisects** the angle: two **equal** halves.

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☞ *Video: Copy a segment*

Key Idea

Set the compass to AB. Put the point on C and draw an arc across the ray. Where it crosses is D.
Now $CD = AB$ — copied without **measuring**.

☞ *Video: Your turn*

Your Turn

You want to bisect a segment that is 8 cm long. Which compass width works: 3 cm or 5 cm?

Answer: 5 **cm**, because it must be more than half of 8 cm, which is 4 **cm**.
With 3 cm the arcs from A and B never **meet**.

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